quantum physics tell us about the world? According to the Copenhagen interpretation, this question has a very simple answer: quantum physics tells us nothing whatsoever about the world.

Rather than telling us a story about the quantum world that atoms and subatomic particles inhabit, the Copenhagen interpretation states that quantum physics is merely a tool for calculating the probabilities of various outcomes of experiments. According to Bohr, there isn't a story about the quantum world because "there is no quantum world. There is only an abstract quantum physical description." That description doesn't allow us to do more than predict probabilities for quantum events, because quantum objects don't exist in the same way as the everyday world around us. As Heisenberg put it, "The idea of an objective real world whose smallest parts exist objectively in the same sense as stones or trees exist, independently of whether or not we observe them, is impossible." But the results of our experiments are very real, because we create them in the process of measuring them. Jordan said when measuring the position of a subatomic particle such as an electron, "the electron is forced to a decision. We compel it *to assume a definite position;* previously, it was, in general, neither here nor there.… We ourselves produce the results of measurement."

Statements like these sounded ludicrous to Albert Einstein. "The theory reminds me a little of the system of delusions of an exceedingly intelligent paranoiac," he said in a letter to a friend. Despite his crucial role in the development of quantum physics, Einstein couldn't stand the Copenhagen interpretation. He called it a "tranquilizing philosophy—or religion" that provides a "soft pillow to the true believer… [but it] has so damned little effect on me." Einstein demanded an interpretation of quantum physics that told a coherent story about the world, one that allowed answers to questions even when no measurement was taking place. He was exasperated with the Copenhagen interpretation's refusal to answer such questions, calling it an "epistemology-soaked orgy."

Yet Einstein's pleas for a more complete theory went unheard, in part because of John von Neumann's proof that no such theory was possible. Von Neumann was arguably the greatest mathematical genius alive. He had taught himself calculus by the age of eight, published his first paper on advanced mathematics at nineteen, and earned a PhD when he was twenty-two. He played a crucial role in building the atomic bomb, and he was one